

Solution Requirements

Date	5 July 2026
Team ID	XXXXXX
Project Name	Emotion Detection & Learning Support Engine
Maximum Marks	4 Marks

Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	<i>Users will securely register and log in using a username and password, which are encrypted and stored in a local SQLite database.</i>	High
2	Authorization levels	<i>Standard Student Role: Can submit learning challenges and view their own analytics history.</i>	Medium
3	External interfaces	<i>The system must connect to the Google Gemini API to generate the personalized, empathetic learning strategies based on predicted emotion</i>	High
4	Transactions processing	<i>The core system processes user text through BiLSTM/BERT models and saves the interaction (emotion, input, AI response) into an SQLite Database and local CSV log.</i>	High

5	Reporting	<i>The system must generate an interactive Analytics Dashboard using Plotly, displaying emotion distribution (Pie charts) and emotional journey timelines (Line charts).</i>	Medium
6	Business rules, etc.	<i>The system requires text inputs to be at least 3 characters long. If the input is too short, the system must reject it and ask for a detailed description.</i>	High
7	Compliance to laws or regulations.	<i>User study data and emotional states must be stored locally to ensure data privacy and prevent sharing of Personally Identifiable Information (PII).</i>	Medium
8	Other	<i>Quick Actions: The system must provide "Quick Example" buttons that auto-fill the text area to help users test the system rapidly</i>	Low

Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	<i>The emotion prediction and Gemini API response generation should be processed quickly so students don't lose focus</i>	<i>Response is generated in < 5 seconds.</i>
2	Scalability	<i>The Streamlit architecture must be able to handle multiple students using the web app at the same time.</i>	<i>Supports up to 50 concurrent users smoothly.</i>
3	Security & Data Privacy	<i>Passwords must not be stored in plain text to protect user accounts from local breaches.</i>	<i>Passwords hashed using standard Python encryption algorithms.</i>
4	Reliability & Availability	<i>The Streamlit web server must run consistently and handle errors gracefully without crashing the app.</i>	<i>99% uptime during active user study sessions.</i>
5	Usability & Accessibility	<i>The UI must be highly intuitive, clean, and not confusing, as the target users are already stressed/frustrated.</i>	<i>Simple navigation, taking < 3 clicks to get an AI response.</i>

6	Other	<i>The Python codebase must be modular so future developers can easily upgrade models or the UI.</i>	Code is cleanly split into app.py, database.py, and predict.py
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